Math 106L In Class 2 Practice Solutions

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Position from Acceleration

1. Suppose that $s(t)$ gives the position of an object moving along a straight line in meters at time t seconds. The object starts 2 meters to the right of a reference point, moving at 3 m/s to the left. The object has a constant acceleration of 2 m/s^2. Find the velocity and position functions and describe the motion.

   Calculus Steps:

   The motion is governed by the second-order differential equation $s''(t) = 2$ m/s^2 with initial conditions $s(0) = 2$ m and $s'(0) = -3$ m/s.

   Antidifferentiate $s''(t) = 2$ to get $s'(t) = v(t) = 2t + C_1$.

   Using the initial condition $v(0) = -3$ m/s, we find $C_1 = -3$, so $v(t) = 2t - 3$ m/s.

   Antidifferentiate $v(t) = 2t - 3$ to get $s(t) = t^2 - 3t + C_2$.

   Using the initial condition $s(0) = 2$ m, we find $C_2 = 2$, so $s(t) = t^2 - 3t + 2$ m.

   Key Values:

   The object stops when $v(t) = 0 \implies 2t - 3 = 0 \implies t = 1.5$ s.

   At this time, the position is $s(1.5) = -0.25$ m.

   The object crosses the reference point when $s(t) = 0 \implies (t-1)(t-2) = 0$, which occurs at $t = 1$ s and $t = 2$ s.

   The velocities at these crossings are $v(1) = -1$ m/s and $v(2) = 1$ m/s.

   Description:

   Initially, the particle travels left for 1.5 seconds. During the interval $0 \le t \le 1.5$, the particle is **slowing down** because the velocity is negative while the acceleration is positive.

   It passes the reference point after 1 second, moving at 1 m/s. At time 1.5 seconds, it comes to a momentary halt 0.25 meters to the left of the reference point.

   For $t \ge 1.5$, the particle is **speeding up** as it moves to the right. It passes the reference point again at time 2 seconds, moving at 1 m/s.

2. A particle starts at the reference point moving at 8 m/s to the right. It is subject to a constant acceleration of -4 m/s^2. Find $s(t)$ and describe the motion.

   Calculus Steps:

   The motion is governed by the second-order differential equation $s''(t) = -4$ m/s^2 with initial conditions $s(0) = 0$ m and $s'(0) = 8$ m/s.

   Antidifferentiate $s''(t) = -4$ to get $s'(t) = v(t) = -4t + C_1$.

   Using the initial condition $v(0) = 8$ m/s, we find $C_1 = 8$, so $v(t) = -4t + 8$ m/s.

   Antidifferentiate $v(t) = -4t + 8$ to get $s(t) = -2t^2 + 8t + C_2$.

   Using the initial condition $s(0) = 0$ m, we find $C_2 = 0$, so $s(t) = -2t^2 + 8t$ m.

   Key Values:

   The object stops when $v(t) = 0 \implies -4t + 8 = 0 \implies t = 2$ s.

   At this time, the position is $s(2) = 8$ m.

   The object returns to the reference point when $s(t) = 0 \implies -2t(t-4) = 0$, which occurs at $t = 0$ s and $t = 4$ s.

   The velocity at the return crossing is $v(4) = -8$ m/s.

Description:

Initially, the particle travels right for 2 seconds. During the interval $0 \leq t \leq 2$, the particle is **slowing down** since it is moving right but accelerating left.

At time 2 seconds, it comes to a momentary halt 8 meters to the right of the reference point.

For $t \geq 2$, the particle is **speeding up** to the left. It passes the reference point again at time 4 seconds, moving at 8 m/s to the left.

3. An object starts 6 meters to the left of a reference point with an initial velocity of 5 m/s to the right. It experiences a constant acceleration of -2 m/s^2. Find $s(t)$ and describe the motion.

Calculus Steps:

The motion is governed by the second-order differential equation $s''(t) = -2$ m/s^2 with initial conditions $s(0) = -6$ m and $s'(0) = 5$ m/s.

Antidifferentiate $s''(t) = -2$ to get $s'(t) = v(t) = -2t + C_1$.

Using the initial condition $v(0) = 5$ m/s, we find $C_1 = 5$, so $v(t) = -2t + 5$ m/s.

Antidifferentiate $v(t) = -2t + 5$ to get $s(t) = -t^2 + 5t + C_2$.

Using the initial condition $s(0) = -6$ m, we find $C_2 = -6$, so $s(t) = -t^2 + 5t - 6$ m.

Key Values:

The object stops when $v(t) = 0 \implies t = 2.5$ s.

At this time, the position is $s(2.5) = 0.25$ m.

The object crosses the reference point when $s(t) = 0 \implies -(t-2)(t-3) = 0$, which occurs at $t = 2$ s and $t = 3$ s.

Description:

Initially, the particle travels right for 2.5 seconds. During the interval $0 \leq t \leq 2.5$, the particle is **slowing down**.

It passes the reference point after 2 seconds, moving at 1 m/s. At time 2.5 seconds, it comes to a momentary halt 0.25 meters to the right of the reference point.

For $t \geq 2.5$, the particle is **speeding up** to the left. It passes the reference point again at time 3 seconds, moving at 1 m/s to the left.

4. Suppose an object is at $s(0) = 3$ meters and $v(0) = -4$ m/s with a constant acceleration of 2 m/s^2. Solve for $s(t)$ and describe the motion.

Calculus Steps:

The motion is governed by the second-order differential equation $s''(t) = 2$ m/s^2 with initial conditions $s(0) = 3$ m and $s'(0) = -4$ m/s.

Antidifferentiate $s''(t) = 2$ to get $s'(t) = v(t) = 2t + C_1$.

Using the initial condition $v(0) = -4$ m/s, we find $C_1 = -4$, so $v(t) = 2t - 4$ m/s.

Antidifferentiate $v(t) = 2t - 4$ to get $s(t) = t^2 - 4t + C_2$.

Using the initial condition $s(0) = 3$ m, we find $C_2 = 3$, so $s(t) = t^2 - 4t + 3$ m.

Key Values:

The object stops when $v(t) = 0 \implies t = 2$ s.

At this time, the position is $s(2) = -1$ m.

The object crosses the reference point when $s(t) = 0 \implies (t-1)(t-3) = 0$, which occurs at $t = 1$ s and $t = 3$ s.

Description:

Initially, the particle travels left for 2 seconds. During the interval $0 \le t \le 2$, the particle is **slowing down**.

It passes the reference point after 1 second, moving at 2 m/s to the left. At time 2 seconds, it comes to a momentary halt 1 meter to the left of the reference point.

For $t \ge 2$, the particle is **speeding up** to the right. It passes the reference point again at time 3 seconds, moving at 2 m/s.

5. An object starts 4 meters to the right of a reference point, moving 4 m/s to the left, with a constant acceleration of 2 m/s^2. Find $s(t)$ and describe the motion.

Calculus Steps:

The motion is governed by the second-order differential equation $s''(t) = 2$ m/s^2 with initial conditions $s(0) = 4$ m and $s'(0) = -4$ m/s.

Antidifferentiate $s''(t) = 2$ to get $s'(t) = v(t) = 2t + C_1$.

Using the initial condition $v(0) = -4$ m/s, we find $C_1 = -4$, so $v(t) = 2t - 4$ m/s.

Antidifferentiate $v(t) = 2t - 4$ to get $s(t) = t^2 - 4t + C_2$.

Using the initial condition $s(0) = 4$ m, we find $C_2 = 4$, so $s(t) = t^2 - 4t + 4$ m.

Key Values:

The object stops when $v(t) = 0 \implies t = 2$ s.

At this time, the position is $s(2) = 0$ m.

Description:

Initially, the particle travels left for 2 seconds. During the interval $0 \le t \le 2$, the particle is **slowing down**.

At time 2 seconds, it comes to a momentary halt exactly at the reference point ($s = 0$ m).

For $t \ge 2$, the particle is **speeding up** as it moves back to the right.

6. A researcher on Earth ($g = 9.8$ m/s^2) stands on the edge of a 50-meter cliff and throws a stone **upward** with an initial speed of 15 m/s. Find the position function $s(t)$, the maximum height reached by the stone, and the total time it takes for the stone to hit the ground at the base of the cliff.

Calculus Steps:

The motion is governed by the second-order differential equation $s''(t) = -9.8$ m/s^2. Since the stone is thrown upward, the initial velocity is positive: $s'(0) = 15$ m/s, and the initial position is $s(0) = 50$ m.

Antidifferentiate $s''(t) = -9.8$ to get $s'(t) = v(t) = -9.8t + C_1$. Using $v(0) = 15$ m/s, we find $C_1 = 15$, so $v(t) = -9.8t + 15$ m/s.

Antidifferentiate $v(t) = -9.8t + 15$ to get $s(t) = -4.9t^2 + 15t + C_2$. Using $s(0) = 50$ m, we find $C_2 = 50$, so $s(t) = -4.9t^2 + 15t + 50$ m.

Maximum Height:

The stone reaches its peak when $v(t) = 0$:

$$-9.8t + 15 = 0 \implies t \approx 1.53 \text{ s}$$

Substitute this into the position function:

$$s(1.53) = -4.9(1.53)^2 + 15(1.53) + 50 \approx 61.48 \text{ m}$$

The maximum height is approximately 61.48 meters.

Total Time to Ground:

The stone hits the ground when $s(t) = 0$:

$$-4.9t^2 + 15t + 50 = 0$$

Using the **quadratic formula**:

$$t = \frac{-15 \pm \sqrt{15^2 - 4(-4.9)(50)}}{2(-4.9)} = \frac{-15 \pm \sqrt{1205}}{-9.8}$$

$t \approx 5.07$ s or $t \approx -2.01$ s. We **reject** $t \approx -2.01$ s because time must be non-negative ($t \geq 0$) as the motion begins at the moment the stone is thrown. The total time in the air is 5.07 seconds.

7. An object is thrown **downward** from the top of a 100-meter building on Earth with an initial speed of 10 m/s. Determine the position function and the velocity of the object at the moment it impacts the ground.

Calculus Steps:

The motion is governed by $s''(t) = -9.8$ m/s^2. Because the object is thrown downward, the initial velocity is negative: $s'(0) = -10$ m/s, with $s(0) = 100$ m.

Antidifferentiate $s''(t) = -9.8$ to get $v(t) = -9.8t - 10$ m/s.

Antidifferentiate $v(t) = -9.8t - 10$ to get $s(t) = -4.9t^2 - 10t + 100$ m.

Impact Velocity:

Find impact time by setting $s(t) = 0$:

$$-4.9t^2 - 10t + 100 = 0$$

Using the **quadratic formula**:

$$t = \frac{10 \pm \sqrt{(-10)^2 - 4(-4.9)(100)}}{2(-4.9)} = \frac{10 \pm \sqrt{2060}}{-9.8}$$

$t \approx 3.61$ s or $t \approx -5.65$ s. We **reject** $t \approx -5.65$ s since the domain for our motion is $t \geq 0$.

The velocity at $t = 3.61$ s is:

$$v(3.61) = -9.8(3.61) - 10 \approx -45.38 \text{ m/s}$$

The impact velocity is 45.38 m/s downward.

8. On Planet X, an astronaut **drops** a heavy tool from a height of 20 meters. The tool takes exactly 4 seconds to hit the surface. Find the value of acceleration due to gravity (g) on Planet X and write the position function $s(t)$.

Calculus Steps:

Let $s''(t) = -a$. Since it is dropped, $s'(0) = 0$ m/s and $s(0) = 20$ m.

Antidifferentiate $s''(t) = -a$ to get $v(t) = -at$ m/s.

Antidifferentiate $v(t) = -at$ to get $s(t) = -0.5at^2 + 20$ m.

Finding Gravity (g):

Use the landing time $s(4) = 0$:

$$0 = -0.5a(4)^2 + 20 \implies 8a = 20 \implies a = 2.5 \text{ m/s}^2$$

The magnitude of gravity on Planet X is 2.5 m/s^2, so $s''(t) = -2.5$ m/s^2. The position function is $s(t) = -1.25t^2 + 20$ m.

Verifying Time:

Setting $s(t) = 0 \implies 1.25t^2 = 20 \implies t^2 = 16$. **Factoring**: $(t-4)(t+4) = 0$. $t = 4$ s or $t = -4$ s. We **reject** $t = -4$ s as it is physically irrelevant.

9. A ball is **dropped** from a height of 30 meters on Earth. How long does it take to reach the halfway point (15 meters)?

Calculus Steps:

$s''(t) = -9.8$ m/s^2, $s'(0) = 0$ m/s, $s(0) = 30$ m.

Antidifferentiate $s''(t) = -9.8$ to get $v(t) = -9.8t$ m/s.

Antidifferentiate $v(t) = -9.8t$ to get $s(t) = -4.9t^2 + 30$ m.

Time to 15 Meters:

Set $s(t) = 15$:

$$-4.9t^2 + 30 = 15 \implies -4.9t^2 = -15 \implies t^2 \approx 3.06$$

$t \approx 1.75$ s or $t \approx -1.75$ s. We **reject** $t \approx -1.75$ s because the ball was not moving prior to $t = 0$. It takes 1.75 seconds to reach the halfway point.

10. On a distant moon, a rock is thrown **upward** at 4 m/s from the surface ($s(0) = 0$). It reaches a maximum height of 5 meters. Find the acceleration due to gravity (g) on this moon and determine the total time the rock is in the air.

Calculus Steps:

$s''(t) = -a$, $s'(0) = 4$ m/s, $s(0) = 0$ m.

Antidifferentiate $s''(t) = -a$ to get $v(t) = -at + 4$ m/s.

Antidifferentiate $v(t) = -at + 4$ to get $s(t) = -0.5at^2 + 4t$ m.

Finding Gravity (g):

Peak height occurs when $v(t) = 0 \implies t = 4/a$. Substitute this into $s(t)$:

$$5 = -0.5a\left(\frac{4}{a}\right)^2 + 4\left(\frac{4}{a}\right) \implies 5 = \frac{-8}{a} + \frac{16}{a} \implies 5 = \frac{8}{a}$$

$a = 1.6$ m/s^2. The magnitude of gravity is 1.6 m/s^2.

Total Time:

The rock returns to the surface when $s(t) = 0$:

$$-0.8t^2 + 4t = 0$$

Factoring: $-0.8t(t-5) = 0$. $t = 0$ s or $t = 5$ s. $t = 0$ s represents the initial throw, so the total time in the air is 5 seconds.

Sums and Sigma Notation

1. Write the following sum in summation notation:

$$\frac{1}{1} + \frac{1}{1.2} + \frac{1}{1.4} + \cdots + \frac{1}{2.4}$$

Identify the Pattern: The numerator is constant at 1. The denominator starts at 1 and increases by a constant gap of 0.2 for each subsequent term.

Find the i-th Term: Let the index i start at 0. The denominator can be expressed as $1 + 0.2i$.

Find the Upper Limit: Set the denominator of the general term equal to the last denominator: $1 + 0.2i = 2.4 \implies 0.2i = 1.4 \implies i = 7$.

Summation Notation:

$$\sum_{i=0}^{7} \frac{1}{1 + 0.2i}$$

2. Write the following sum in summation notation:

$$(0.5)^2 + (1.0)^2 + (1.5)^2 + \cdots + (10)^2$$

Identify the Pattern: The base of the power starts at 0.5 and increases by a constant gap of 0.5. Each base is then squared.

Find the i-th Term: Let i start at 1. the base is $0.5i$.

Find the Upper Limit: Set the base equal to the last base: $0.5i = 10 \implies i = 20$.

Summation Notation:

$$\sum_{i=1}^{20} (0.5i)^2$$

3. Write the following sum in summation notation:

$$e^3 + e^{3.25} + e^{3.5} + e^{3.75} + \cdots + e^6$$

Identify the Pattern: The base e is constant. The exponents start at 3 and increase by a constant gap of 0.25.

Find the i-th Term: Let i start at 0. The exponent is $3 + 0.25i$.

Find the Upper Limit: $3 + 0.25i = 6 \implies 0.25i = 3 \implies i = 12$.

Summation Notation:

$$\sum_{i=0}^{12} e^{3+0.25i}$$

4. Write the following sum in summation notation:

$$\ln(2.1) + \ln(2.2) + \ln(2.3) + \cdots + \ln(3.5)$$

Identify the Pattern: The argument of the natural logarithm starts at 2.1 and increases by a constant gap of 0.1 per term.

Find the i-th Term: Let i start at 0. The argument is $2.1 + 0.1i$.

Find the Upper Limit: $2.1 + 0.1i = 3.5 \implies 0.1i = 1.4 \implies i = 14.$

Summation Notation:

$$\sum_{i=0}^{14} \ln(2.1 + 0.1i)$$

5. Write the following sum in summation notation:

$$\sqrt{5} + \sqrt{5.05} + \sqrt{5.1} + \cdots + \sqrt{6}$$

Identify the Pattern: The value under the square root starts at 5 and increases by a constant gap of 0.05.

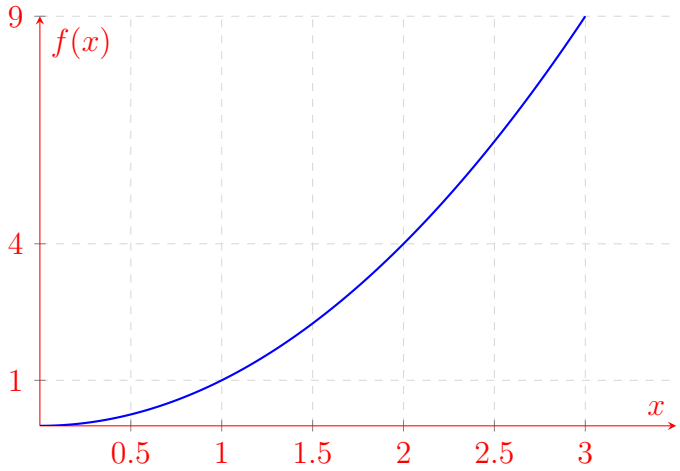
Find the i-th Term: Let i start at 0. The radicand is $5 + 0.05i$.

Find the Upper Limit: $5 + 0.05i = 6 \implies 0.05i = 1 \implies i = 20.$

Summation Notation:

$$\sum_{i=0}^{20} \sqrt{5 + 0.05i}$$

6. Consider the function $f(x) = x^2$ on the interval $[0, 2]$ with $n = 4$ subintervals.



(a) On the graph above, draw the **right-hand** rectangles used to estimate the area under the curve.

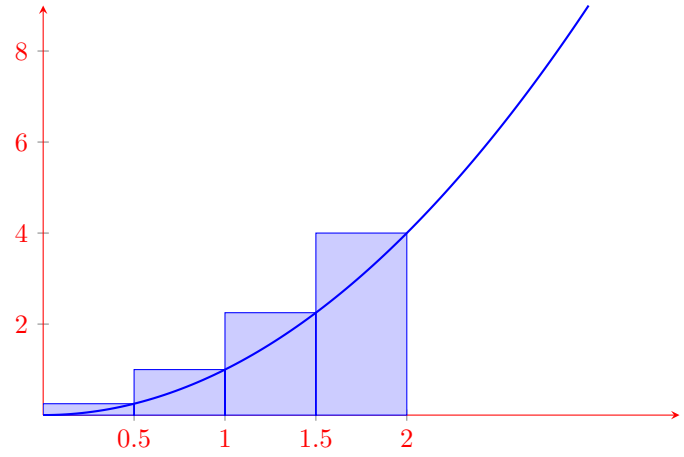
(b) Write down the sequence of the right endpoints of the subintervals.

(c) Write down the sequence of heights of the right-hand rectangles.

(d) Write down the right-hand sum in sigma notation.

Calculus Steps: The width of each subinterval is $\Delta x = \frac{2-0}{4} = 0.5$.

a) Drawing:

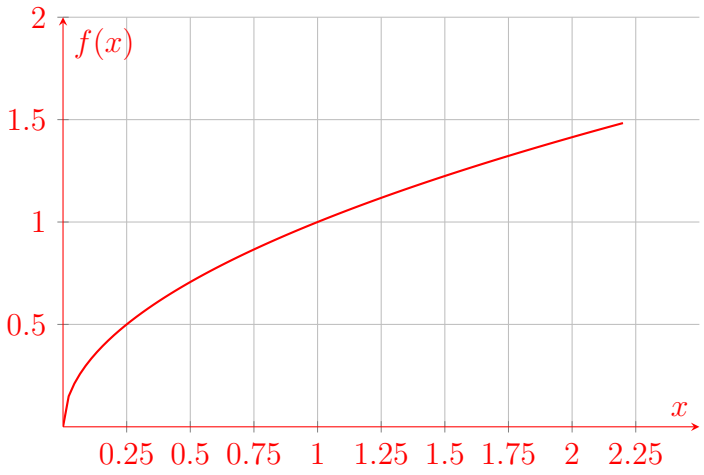


b) Right Endpoints: $0.5, 1.0, 1.5, 2.0$.

c) Heights: $f(0.5) = 0.25,\ f(1.0) = 1,\ f(1.5) = 2.25,\ f(2.0) = 4.$

d) Sigma Notation: $\sum_{i=1}^{4} (0.5i)^2 \cdot 0.5$

7. Consider the function $f(x) = \sqrt{x}$ on the interval $[0, 1.5]$ with $n = 6$ subintervals.



(a) On the graph above, draw the **left-hand** rectangles used to estimate the area under the curve.

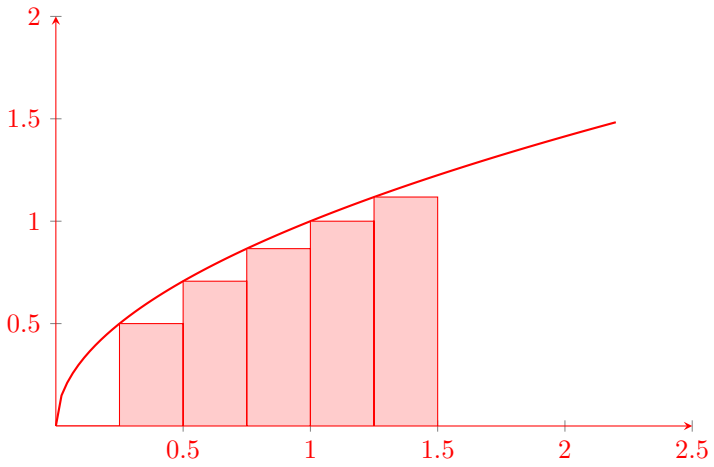
(b) Write down the sequence of the left endpoints of the subintervals.

(c) Write down the sequence of heights of the left-hand rectangles.

(d) Write down the left-hand sum in sigma notation.

Calculus Steps: The width of each subinterval is $\Delta x = \frac{1.5-0}{6} = 0.25$.

a) Drawing:



b) Left Endpoints: $0, 0.25, 0.5, 0.75, 1, 1.25$.

c) Heights: $\sqrt{0}, \sqrt{0.25}, \sqrt{0.5}, \sqrt{0.75}, \sqrt{1}, \sqrt{1.25}$.

d) Sigma Notation: $\sum_{i=0}^{5} \sqrt{0.25i} \cdot 0.25$

8. Consider $f(x) = \frac{1}{x}$ on the interval $[1, 3]$ with $n = 50$ subintervals.

   (a) Write down the sequence of the right endpoints of the subintervals.

   (b) Write down the sequence of heights of the right-hand rectangles.

   (c) Write down the right-hand sum in sigma notation.

Calculus Steps: The width is $\Delta x = \frac{3-1}{50} = 0.04$.

a) Right Endpoints: $1.04, 1.08, 1.12, \ldots, 3.0$.

b) Heights: $\frac{1}{1.04}, \frac{1}{1.08}, \frac{1}{1.12}, \ldots, \frac{1}{3}$.

c) Sigma Notation: $\sum_{i=1}^{50} \frac{1}{1 + 0.04i} \cdot 0.04$

9. Consider $f(x) = \sin(x)$ on $[0, \pi]$ with $n = 100$ subintervals.

   (a) Write down the sequence of the left endpoints of the subintervals.

   (b) Write down the sequence of heights of the left-hand rectangles.

   (c) Write down the left-hand sum in sigma notation.

Calculus Steps: The width is $\Delta x = \frac{\pi-0}{100} = 0.01\pi$.

a) Left Endpoints: $0, 0.01\pi, 0.02\pi, \ldots, 0.99\pi$.

b) Heights: $\sin(0), \sin(0.01\pi), \sin(0.02\pi), \ldots, \sin(0.99\pi)$.

c) Sigma Notation: $\sum_{i=0}^{99} \sin(0.01\pi i) \cdot 0.01\pi$

10. Consider $f(x) = e^x$ on $[0, 5]$ with $n = 200$ subintervals.

    (a) Write down the sequence of the right endpoints of the subintervals.

    (b) Write down the sequence of heights of the right-hand rectangles.

(c) Write down the right-hand sum in sigma notation.

Calculus Steps: The width is $\Delta x = \frac{5-0}{200} = 0.025$.

a) Right Endpoints: $0.025, 0.05, 0.075, \ldots, 5.0$.

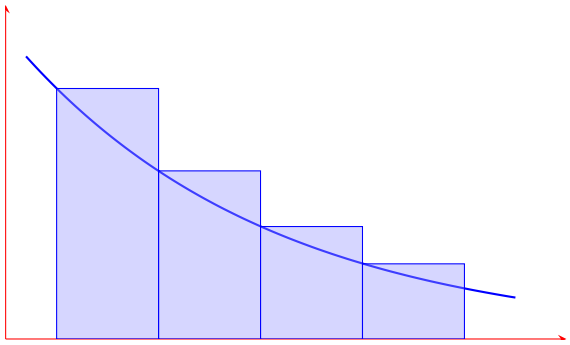
b) Heights: $e^{0.025}, e^{0.05}, e^{0.075}, \ldots, e^{5}$.

c) Sigma Notation: $\displaystyle\sum_{i=1}^{200} e^{0.025i} \cdot 0.025$

Left and Right Hand Sums

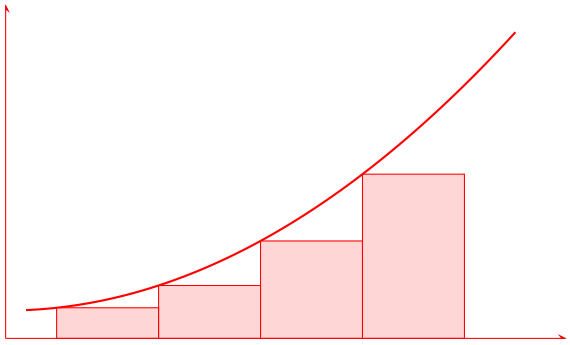
1. **Fill in the blanks**: A Left Hand Sum overestimates the integral of $f(x)$ when the function is decreasing. That is, when $f'(x) < 0$. Draw a graph and rectangles on it to illustrate your answer.

   Explanation: On a decreasing interval, the left endpoint of each subinterval represents the maximum value for that segment. Because the rectangle height is fixed at this "high point," the area of the rectangle extends above the actual curve.



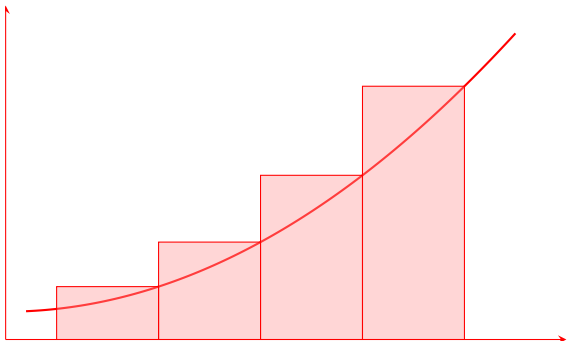
2. **Fill in the blanks**: A Left Hand Sum underestimates the integral of $f(x)$ when the function is increasing. That is, when $f'(x) > 0$. Draw a graph and rectangles on it to illustrate your answer.

   Explanation: On an increasing interval, the left endpoint provides the minimum value for the subinterval. Since the rectangle's height is pinned to this lowest point, it leaves a gap of uncounted area between the top of the rectangle and the rising curve.



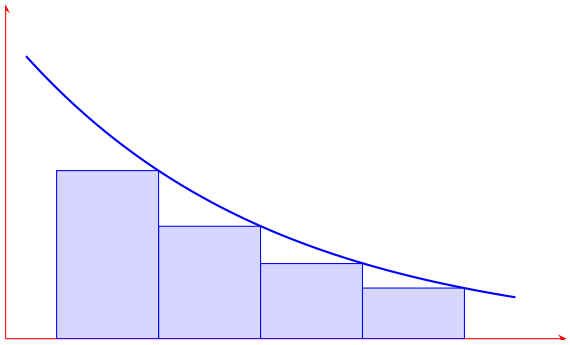
3. **Fill in the blanks**: A Right Hand Sum overestimates the integral of $f(x)$ when the function is increasing. That is, when $f'(x) > 0$. Draw a graph and rectangles on it to illustrate your answer.

   Explanation: For an increasing function, the right endpoint is the maximum value on any given subinterval. The height of the rectangle therefore encompasses all the area under the curve plus an extra region of space that sits above the curve.



4. **Fill in the blanks**: A Right Hand Sum underestimates the integral of $f(x)$ when the function is decreasing. That is, when $f'(x) < 0$. Draw a graph and rectangles on it to illustrate your answer.

   Explanation: For a decreasing function, the right endpoint is the minimum value on the subinterval. Because the rectangle's height is determined by this "low point," it stays entirely beneath the curve, missing the higher area found at the start of the interval.



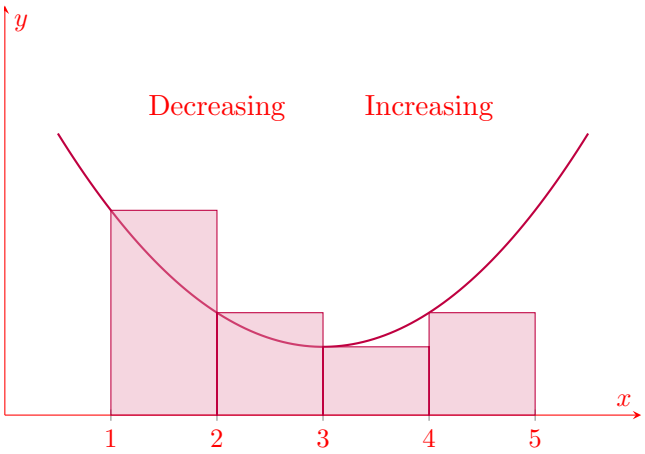
5. Consider $f(x) = \frac{1}{x}$ on the interval $[1, 5]$. Does a Right Hand Sum underestimate or overestimate the integral? Show any needed computations.

   Calculus Check: Since $f'(x) = -1/x^2$, the derivative is strictly negative for all x in $[1, 5]$. Because the function is decreasing, the Right Hand Sum uses the lowest point of each sub-interval as the height, resulting in a sum smaller than the true area.

6. Draw a curve $f(x)$ on the interval $[1, 5]$ such that a Left Hand Sum overestimates the integral on the interval $[1, 3]$ and underestimates the integral on the interval $[3, 5]$.

   Explain what property of the function changes at $x = 3$ to cause this shift in estimation behavior. Draw rectangles on your graph to illustrate this.

   Explanation: At $x = 3$, the function changes from **decreasing** to **increasing**. This indicates a local minimum. On $[1, 3]$, $f'(x) < 0$ makes the left endpoint the highest point (overestimate). On $[3, 5]$, $f'(x) > 0$ makes the left endpoint the lowest point (underestimate).



7. Suppose that $f(x)$ is an decreasing, continuous function. Write the following quantities from least to greatest:

$$\text{LHS}_5 \qquad \text{LHS}_{10} \qquad \text{RHS}_5 \qquad \text{RHS}_{10}$$

Explain your answer.

$\text{RHS}_5 < \text{RHS}_{10} < \text{LHS}_{10} < \text{LHS}_5$.

Since RHS is an underestimate of a decreasing function, and LHS an overestimate, we know that all RHS's are less than all LHS's.

Making an underestimate better means getting it larger. Since RHS_10 is a better underestimate than RHS_5, it is larger.

Making an overestimate better means getting it smaller. Since LHS_10 is a better overestimate than LHS_5, it is smaller.

8. Write down the definition of the definite integral of $f(x)$ from $x = a$ to $x = b$ using a left-hand sum.

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(a + (i-1)\Delta x)\Delta x$$

where $\Delta x = \frac{b-a}{n}$.

9. Explain why it is is *not true* in general that

$$\sum_{i=1}^{n} f(a + (i-1)\Delta x)\Delta x = \sum_{i=1}^{n} f(a + i\Delta x)\Delta x$$

but it is true that

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(a + (i-1)\Delta x)\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} f(a + i\Delta x)\Delta x$$

where $\Delta x = \frac{b-a}{n}$.

The first line is LHS_n and RHS_n for $f(x)$. Unless the function is constant (neither increasing nor decreasing), these are not equal.

However, by taking the limits as $n \to \infty$, both sums approach the integral $\int_a^b f(x)\,dx$, so they must be the same.

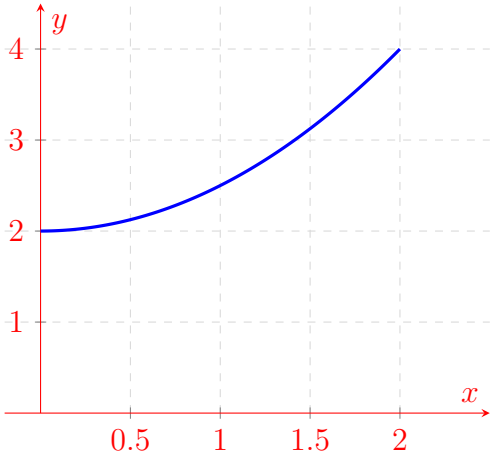
10. Consider the function $f(x) = \frac{1}{2}x^2 + 2$ on the interval $[0, 2]$.

    (a) Divide the interval $[0, 2]$ into $n = 4$ subintervals of equal width. Determine the value of Δx and the endpoints of the subintervals.

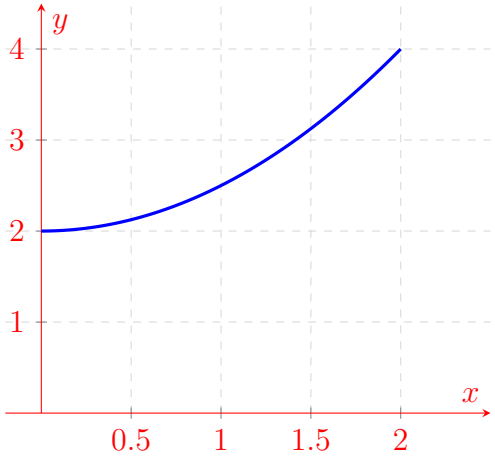
    (b) On **Graph 1**, sketch and shade the rectangles used to calculate the **Left-Hand Sum** (L_4). Compute the value of LHS_4.

    (c) On **Graph 2**, sketch and shade the rectangles used to calculate the **Right-Hand Sum** (R_4). Compute the value of RHS_4.

Graph 1: Left-Hand Sum



Graph 2: Right-Hand Sum



a) Setup:

$$\Delta x = \frac{2-0}{4} = 0.5$$

Endpoints of intervals: 0, 0.5, 1, 1.5, 2

b) Left-Hand Sum (LHS_4): Uses left endpoints: $\{0, 0.5, 1, 1.5\}$.

- $f(0) = 0.5(0)^2 + 2 = 2$
- $f(0.5) = 0.5(0.25) + 2 = 2.125$
- $f(1) = 0.5(1)^2 + 2 = 2.5$
- $f(1.5) = 0.5(2.25) + 2 = 3.125$

$$\text{LHS}_4 = (2 + 2.125 + 2.5 + 3.125) \cdot 0.5 = 9.75 \cdot 0.5 = 4.875$$

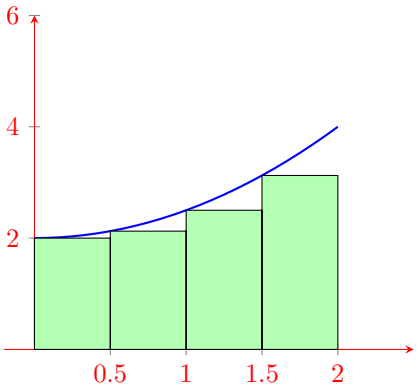
c) Right-Hand Sum (RHS_4): Uses right endpoints: $\{0.5, 1, 1.5, 2\}$.

- $f(0.5) = 2.125$
- $f(1) = 2.5$
- $f(1.5) = 3.125$

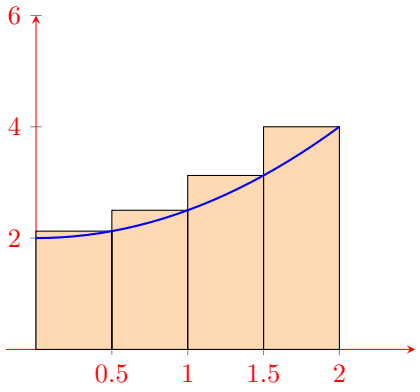
- $f(2) = 0.5(4) + 2 = 4$

$$\text{RHS}_4 = (2.125 + 2.5 + 3.125 + 4) \cdot 0.5 = 11.75 \cdot 0.5 = 5.875$$

LHS Solution

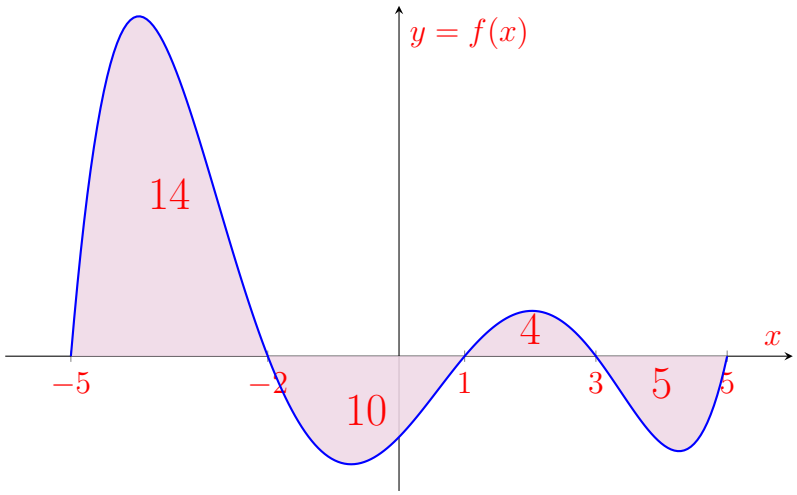


RHS Solution



From Riemann Sums to Definite Integrals

1. The graph of $y = f(x)$ is shown below for the interval $[-5, 5]$. The areas of the four regions bounded by the curve and the x-axis are indicated.



Use the graph and the labeled areas to compute the following definite integrals:

(a) $\displaystyle\int_{-5}^{1} f(x)\,dx$

(b) $\displaystyle\int_{-2}^{3} f(x)\,dx$

(c) $\displaystyle\int_{1}^{5} f(x)\,dx$

(d) $\displaystyle\int_{-5}^{5} f(x)\,dx$

Calculus Steps: Recall that the definite integral is the signed sum of areas: positive for regions above the x-axis and negative for regions below.

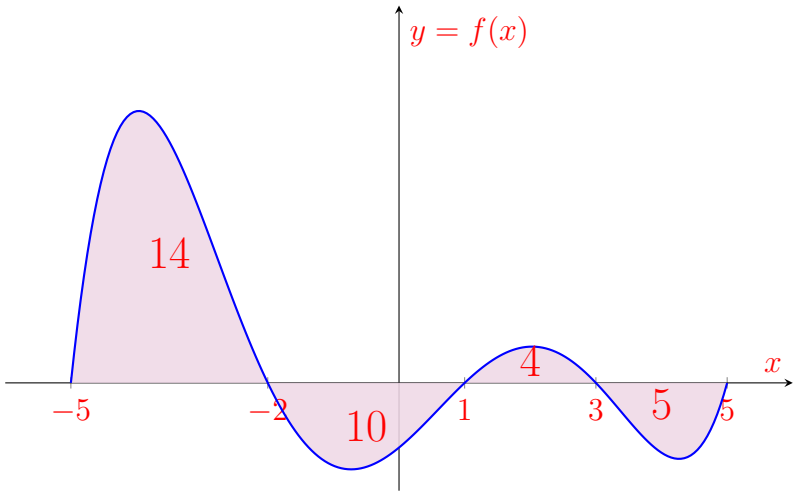
a) $\displaystyle\int_{-5}^{1} f(x)\,dx = 14 - 10 = 4$

b) $\displaystyle\int_{-2}^{3} f(x)\,dx = -10 + 4 = -6$

c) $\displaystyle\int_{1}^{5} f(x)\,dx = 4 - 5 = -1$

d) $\displaystyle\int_{-5}^{5} f(x)\,dx = 14 - 10 + 4 - 5 = 3$

2. The graph of $y = f(x)$ is shown below for the interval $[-5, 5]$. The areas of the four regions bounded by the curve and the x-axis are indicated.



Use the graph and the labeled areas to compute the following definite integrals:

(a) $\int_{-5}^{1} f(x)\,dx$

(b) $\int_{-2}^{3} f(x)\,dx$

(c) $\int_{1}^{5} f(x)\,dx$

(d) $\int_{-5}^{5} f(x)\,dx$

Calculus Steps: Recall that the definite integral is the signed sum of areas: positive for regions above the x-axis and negative for regions below.

a) $\int_{-5}^{1} f(x)\,dx = 14 - 10 = 4$

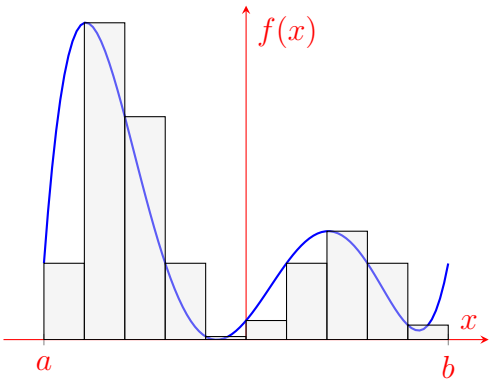
b) $\int_{-2}^{3} f(x)\,dx = -10 + 4 = -6$

c) $\int_{1}^{5} f(x)\,dx = 4 - 5 = -1$

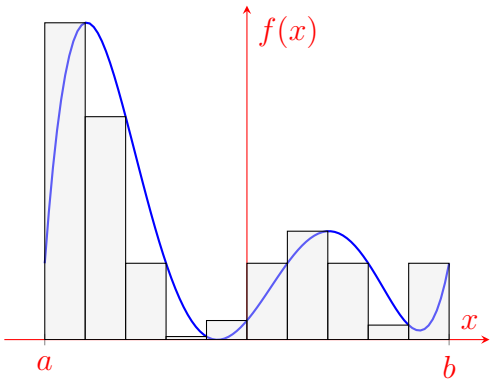
d) $\int_{-5}^{5} f(x)\,dx = 14 - 10 + 4 - 5 = 3$

3. Below are two copies of the graph of the same function $y = f(x)$ on the interval $[a, b]$. Each graph shows a Riemann sum approximation with $n = 10$ subintervals of equal width Δx.

Graph 1



Graph 2



(a) Identify which graph represents the **Left-Hand Sum** and which represents the **Right-Hand Sum**.

(b) On **both** graphs, locate and shade in the specific rectangle whose area is given by the term:

$$f(a+5\Delta x)\Delta x$$

a) Identification:

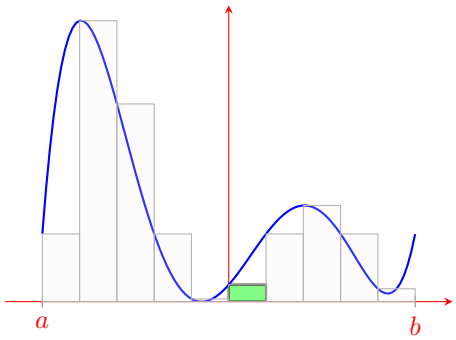
- **Graph 1** is the **Left-Hand Sum** (LHS).
- **Graph 2** is the **Right-Hand Sum** (RHS).

b) Shaded Rectangles:

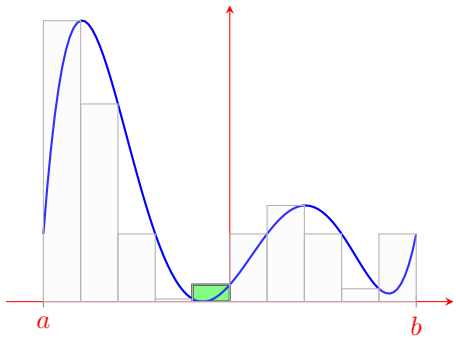
A LHS has the area of the ith rectangle equal to $f(a+(i-1)\Delta x)\Delta x$, so here we want $(i-1)=5$, or $i=6$. Therefore the 6th rectangle in an LHS has this area.

A RHS has the area of the ith rectangle equal to $f(a+i\Delta x)\Delta x$, so here we want $i=5$. Therefore the 5th rectangle in an RHS has this area.

LHS: 6th rectangle area $= f(a+5\Delta x)\Delta x$



RHS: 5th rectangle area $= f(a+5\Delta x)\Delta x$

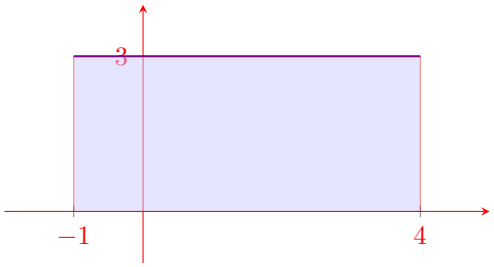


LHS/RHS Precision and Geometric Integrals

1. **Constant Function:** Compute $\int_{-1}^{4} 3\,dx$ using geometry. Sketch the graph and shade the relevant region.

   Geometric Shape: This is a rectangle with height 3.

   - **Width:** $4-(-1)=5$
   - **Height:** 3
   - **Area Calculation:** $5\times 3=15$

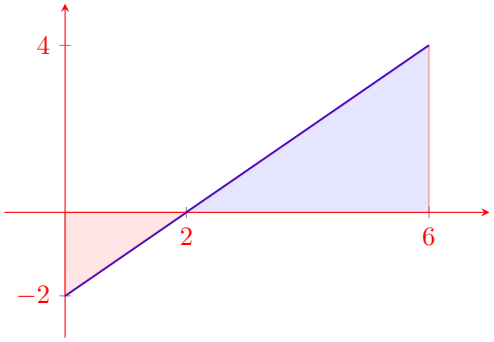


   $\int_{-1}^{4} 3\,dx = 15.$

2. **Linear Function:** Compute $\int_{0}^{6} (x-2)\,dx$ using geometry. Sketch the graph and shade the relevant regions.

   Geometric Shape: Two triangles—one below the axis and one above.

   - **Triangle 1 (Below, $x=0$ to 2):** Base $=2$, Height $=|f(0)|=2$. Area $=\frac{1}{2}(2)(2)=2$. (Signed area $=-2$)
   - **Triangle 2 (Above, $x=2$ to 6):** Base $=4$, Height $=f(6)=4$. Area $=\frac{1}{2}(4)(4)=8$. (Signed area $=+8$)
   - **Net Integral:** $-2+8=6$.

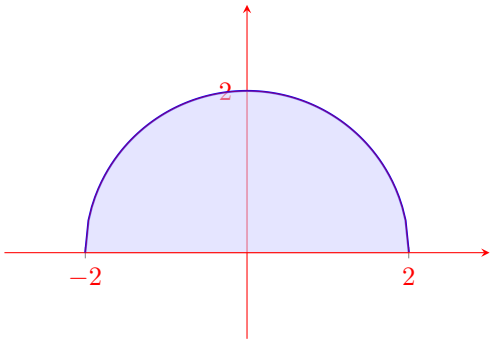


   $\int_{0}^{6} (x-2)\,dx = 6.$

3. **Semicircle:** Compute $\int_{-2}^{2} \sqrt{4-x^2}\,dx$ using geometry. Sketch the graph and shade the relevant region.

   Geometric Shape: A semicircle with radius $r=2$.

- **Radius:** 2
- **Area Calculation:** $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi$.



$\int_{-2}^{2} \sqrt{4-x^2}\,dx = 2\pi$.

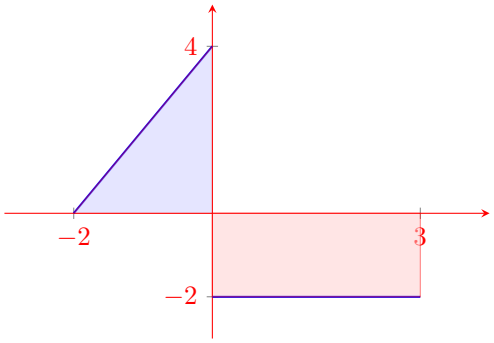
4. **Piecewise Mixed:** Compute $\int_{-2}^{3} f(x)\,dx$ where

$$f(x) = \begin{cases} 2x+4 & -2 \le x \le 0 \\ -2 & 0 < x \le 3 \end{cases}$$

Sketch the graph and shade the regions.

Geometric Shapes: A triangle above the axis and a rectangle below.

- **Triangle (Above, -2 to 0):** Base $= 2$, Height $= f(0) = 4$. Area $= \frac{1}{2}(2)(4) = 4$.
- **Rectangle (Below, 0 to 3):** Width $= 3$, Height $= |-2| = 2$. Area $= 3 \times 2 = 6$. (Signed area $= -6$)
- **Net Integral:** $4 - 6 = -2$.



$\int_{-2}^{3} f(x)\,dx = -2$.

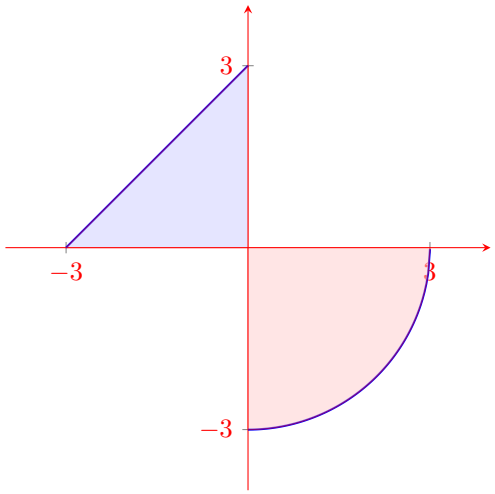
5. **Piecewise Mixed:** Compute $\int_{-3}^{3} g(x)\,dx$ where

$$g(x) = \begin{cases} x+3 & -3 \le x \le 0 \\ -\sqrt{9-x^2} & 0 < x \le 3 \end{cases}$$

Sketch the graph and shade the regions.

Geometric Shapes: A triangle above the axis and a quarter-circle below.

- **Triangle (Above, -3 to 0):** Base $= 3$, Height $= 3$. Area $= \frac{1}{2}(3)(3) = 4.5$.
- **Quarter-Circle (Below, 0 to 3):** A circle centered at $(0,0)$ with radius $r = 3$. Area $= \frac{1}{4}\pi(3^2) = \frac{9\pi}{4}$. (Signed area $= -\frac{9\pi}{4}$)
- **Net Integral:** $4.5 - \frac{9\pi}{4}$.



$\int_{-3}^{3} g(x)\,dx = 4.5 - \frac{9\pi}{4}$.

6. Let $f(x) = 2x + 7$ on the interval $[1, 6]$. Find the minimum number of equal-width subintervals n such that the difference between the Right Hand Sum and the Left Hand Sum is less than 0.01.

   1. Identify values: $f(1) = 9$, $f(6) = 19$, and $b - a = 5$.

   2. Set up inequality:

   $$|f(b) - f(a)|\Delta x = |19 - 9| \cdot \frac{5}{n} < 0.01$$

   $$10 \cdot \frac{5}{n} < 0.01 \implies \frac{50}{n} < 0.01$$

   3. Solve for n:

   $$n > \frac{50}{0.01} = 5000$$

   At least **5001** subintervals are required.

7. Let $f(x) = e^x$ on the interval $[0, 3]$. How many subintervals n are needed so that $|RHS_n - LHS_n| \leq 0.1$? (Use $e^3 \approx 20.086$).

   1. Identify values: $f(0) = 1$, $f(3) = e^3 \approx 20.086$, and $b - a = 3$.

   2. Set up inequality:

   $$|f(b) - f(a)|\Delta x = |20.086 - 1| \cdot \frac{3}{n} \leq 0.1$$

   $$19.086 \cdot \frac{3}{n} \leq 0.1 \implies \frac{57.258}{n} \leq 0.1$$

   3. Solve for n:

   $$n \geq \frac{57.258}{0.1} = 572.58$$

   At least **573** subintervals are required.

8. Let $f(x) = \sin(x)$ on the interval $[0, \pi/2]$. Find n such that $|RHS_n - LHS_n| < 0.05$.

1. Identify values: $f(0) = 0$, $f(\pi/2) = 1$, and $b - a = \pi/2 \approx 1.571$.

2. Set up inequality:

$$|f(b) - f(a)|\Delta x = |1 - 0| \cdot \frac{1.571}{n} \leq 0.05 \implies \frac{1.571}{n} \leq 0.05$$

3. Solve for n:

$$n \geq \frac{1.571}{0.05} = 31.42$$

At least **32** subintervals are required.

9. Let $f(x) = \frac{10}{x}$ on the interval $[2, 10]$. Find n such that $|RHS_n - LHS_n| < 0.25$.

1. Identify values: $f(2) = 5$, $f(10) = 1$, and $b - a = 8$.

2. Set up inequality:

$$|f(b) - f(a)|\Delta x = |1 - 5| \cdot \frac{8}{n} < 0.25$$

$$4 \cdot \frac{8}{n} < 0.25 \implies \frac{32}{n} < 0.25$$

3. Solve for n:

$$n > \frac{32}{0.25} = 128$$

At least **129** subintervals are required.

10. Let $f(x) = \sqrt{x}$ on the interval $[0, 100]$. Find n such that $|RHS_n - LHS_n| < 1$.

1. Identify values: $f(0) = 0$, $f(100) = 10$, and $b - a = 100$.

2. Set up inequality:

$$|f(b) - f(a)|\Delta x = |10 - 0| \cdot \frac{100}{n} < 1 \implies \frac{1000}{n} < 1$$

3. Solve for n:

$$n > 1000$$

At least **1001** subintervals are required.

Midpoint and Trapezoid Sums

1. Consider $\int_0^4 2x+3\,dx$. Compute MPS_2 and TRAP_2. Sketch the rectangles for MPS_2. Explain why MPS and TRAP both give exact answers here.

   Setup: $\Delta x = \frac{4-0}{2} = 2$. Endpoints: $x = (0, 2, 4)$.

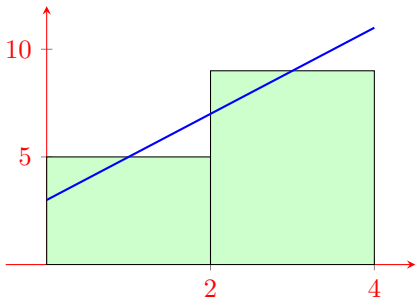
   TRAP_2 Calculation:

   - $\text{LHS}_2 = (f(0) + f(2)) \cdot 2 = (3 + 7) \cdot 2 = 20$
   - $\text{RHS}_2 = (f(2) + f(4)) \cdot 2 = (7 + 11) \cdot 2 = 36$
   - $\text{TRAP}_2 = \frac{20+36}{2} = 28$

   MPS_2 Calculation: Midpoints are 1 and 3.

   $$\text{MPS}_2 = (f(1) + f(3)) \cdot 2 = (5 + 9) \cdot 2 = 28$$

   MPS_2 Sketch

   These answers are exact because our function is neither concave up nor down.

2. Consider $\int_0^2 x^2+1\,dx$. Compute MPS_2 and TRAP_2. Sketch the rectangles for MPS_2. Does MPS overestimate this integral or underestimate it? Why?

   Setup: $\Delta x = 1$. Endpoints: $x = (0, 1, 2)$.

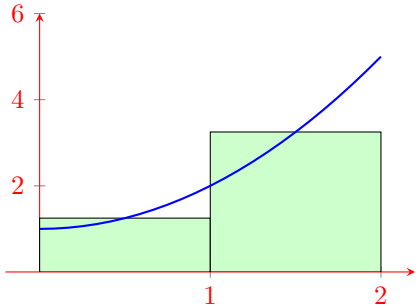
   TRAP_2 Calculation:

   - $\text{LHS}_2 = (f(0) + f(1)) \cdot 1 = (1 + 2) \cdot 1 = 3$
   - $\text{RHS}_2 = (f(1) + f(2)) \cdot 1 = (2 + 5) \cdot 1 = 7$
   - $\text{TRAP}_2 = \frac{3+7}{2} = 5$

   MPS_2 Calculation: Midpoints are 0.5 and 1.5.

   $$\text{MPS}_2 = (f(0.5) + f(1.5)) \cdot 1 = (1.25 + 3.25) = 4.5$$

   MPS_2 Sketch

MPS underestimates this integral since the function is concave up.

3. Consider $\int_1^5 \frac{12}{x}\,dx$. Compute MPS_2 and TRAP_2. Sketch the rectangles for MPS_2. Does TRAP overestimate this integral or underestimate it? Why?

Setup: $\Delta x = 2$. Endpoints: $x = (1, 3, 5)$.

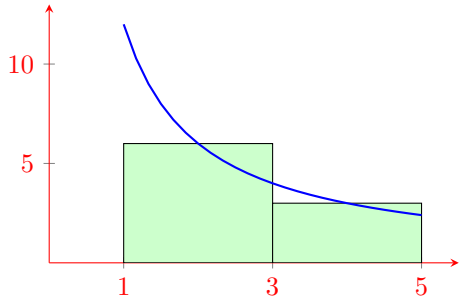
TRAP$_2$ Calculation:

- $\text{LHS}_2 = (f(1) + f(3)) \cdot 2 = (12 + 4) \cdot 2 = 32$
- $\text{RHS}_2 = (f(3) + f(5)) \cdot 2 = (4 + 2.4) \cdot 2 = 12.8$
- $\text{TRAP}_2 = \frac{32+12.8}{2} = 22.4$

MPS$_2$ Calculation: Midpoints are 2 and 4.

$$\text{MPS}_2 = (f(2) + f(4)) \cdot 2 = (6 + 3) \cdot 2 = 18$$

MPS_2 Sketch



TRAP overestimates this integral since the function is concave up.

4. Consider $\int_0^2 10 - e^x\,dx$. Compute MPS_2 and TRAP_2 to three decimal places. Sketch the rectangles for MPS_2. Does MPS overestimate or underestimate this integral? Why?

Setup: $\Delta x = 1$. Endpoints: $x = (0, 1, 2)$.

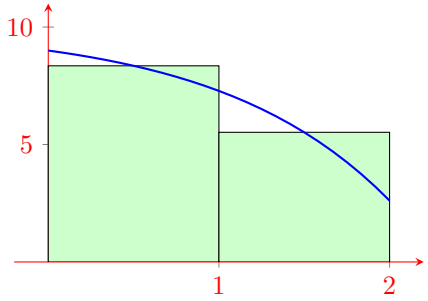
TRAP$_2$ Calculation:

- $\text{LHS}_2 = (f(0) + f(1)) \cdot 1 = 9 + 7.28 = 16.28$
- $\text{RHS}_2 = (f(1) + f(2)) \cdot 1 = 7.28 + 2.61 = 9.89$
- $\text{TRAP}_2 = \frac{16.28+9.89}{2} = 13.085$

MPS$_2$ Calculation: Midpoints are 0.5 and 1.5.

$$\text{MPS}_2 = (f(0.5) + f(1.5)) \cdot 1 = 8.35 + 5.52 = 13.87$$

MPS_2 Sketch



Concave down function, so MPS overestimates.

5. A student wrote the following:

"Since $f(x) = x^2$ is increasing when $x > 0$, a midpoint sum estimate of the integral $\int_2^9 x^2\,dx$ underestimates the integral."

Is their argument correct? If so, explain why. If not, correct it.

While the student is correct that MPS underestimates the integral, it is not because the function is increasing, but rather because it is concave up.

Fundamental Theorem of Calculus

1. Which of the following concepts were involved in our proof of the Fundamental Theorem of Calculus? Circle all correct answers:

   [Linear Approximation] Concavity Critical Points [Riemann Sums]
   [Definite Integrals] Indefinite Integrals [Antiderivatives]

2. The Fundamental Theorem only applies to one of the following two integrals. Compute the integral it applies to, and explain why it does not apply to the other:

$$\int_1^4 \frac{1}{(x-2)^2}\,dx \qquad \int_3^4 \frac{1}{(x-2)^2}\,dx$$

   The function is not continuous at $x = 2$, so the FTC doesn't apply to the first integral, whose domain includes $x = 2$.

$$\int_3^4 \frac{1}{(x-2)^2}\,dx = -\frac{1}{x-2}\Big|_3^4 = -\frac{1}{2} - \left(-\frac{1}{1}\right) = -\frac{1}{2} + 1 = \frac{1}{2}$$

3. The Fundamental Theorem applies to both the following two integrals:

$$\int_3^4 e^x\,dx \qquad \int_3^4 e^{x^2}\,dx$$

   (a) Explain why it applies to both.

   (b) Explain why, despite that, one of these can't actually be computed using the FTC.

   (c) Compute the integral the FTC applies to.

   (a) The functions are both continuous on their domains, so FTC applies.

   (b) e^{x^2} has no computable antiderivative, so its integral can't actually be computed using the FTC.

   (c) $\displaystyle\int_3^4 e^x\,dx = e^x\Big|_3^4 = e^4 - e^3.$

4. Explain why, while the FTC applies to the integral

$$\int_0^4 \sqrt{16 - x^2}\,dx,$$

   and could be used, we don't actually need to use FTC to compute this integral. Then compute the integral exactly.

   This function is continuous on its domain. It has a (complicated) antiderivative too. But the graph is just a quarter circle of radius 4, so the integral is

$$\frac{\pi 4^2}{4} = 4\pi.$$

Integration

1. $\displaystyle\int \frac{\sin(\ln x)}{x}\,dx$

   u-sub.
   Let $u = \ln x$, so $du = \frac{1}{x}dx$.

   $$\int \frac{\sin(\ln x)}{x}\,dx = \int \sin u\,du = -\cos u + C = -\cos(\ln x) + C$$

   Check: $\frac{d}{dx}(-\cos(\ln x)) = -(-\sin(\ln x)) \cdot \frac{1}{x} = \frac{\sin(\ln x)}{x}$.

2. $\displaystyle\int x\cos(3x)\,dx$

   Integration by Parts.

$u = x$	$dv = \cos(3x)dx$
$du = dx$	$v = \frac{1}{3}\sin(3x)$

   $$\int x\cos(3x)\,dx = \frac{1}{3}x\sin(3x) - \int \frac{1}{3}\sin(3x)\,dx = \frac{1}{3}x\sin(3x) + \frac{1}{9}\cos(3x) + C$$

   Check: $\frac{d}{dx}(\frac{1}{3}x\sin(3x) + \frac{1}{9}\cos(3x)) = \frac{1}{3}\sin(3x) + x\cos(3x) - \frac{1}{3}\sin(3x) = x\cos(3x)$.

3. $\displaystyle\int \sin^3 w\,dw$

   (Hint: Use the identity $\sin^2 x + \cos^2 x = 1$ to rewrite the integral before computing)

   Trig Identity + u-sub.
   $\int \sin^2 w \sin w\,dw = \int (1-\cos^2 w)\sin w\,dw$. Let $u = \cos w, du = -\sin w dw$.

   $$-\int (1-u^2)\,du = \int (u^2-1)\,du = \frac{1}{3}u^3 - u + C = \frac{1}{3}\cos^3 w - \cos w + C$$

   Check: $\frac{d}{dw}(\frac{1}{3}\cos^3 w - \cos w) = \cos^2 w(-\sin w) - (-\sin w) = \sin w(1-\cos^2 w) = \sin^3 w$.

4. $\displaystyle\int_0^1 xe^{-x}\,dx$

   Integration by Parts.

$u = x$	$dv = e^{-x}dx$
$du = dx$	$v = -e^{-x}$

   $$-xe^{-x}\Big|_0^1 + \int_0^1 e^{-x}\,dx = (-e^{-1} - 0) + (-e^{-x}\Big|_0^1) = -e^{-1} - e^{-1} + 1 = 1 - 2e^{-1}$$

5. $\displaystyle\int \frac{e^x}{1+e^{2x}}\,dx$

   u-sub (Inverse Trig type).
   Let $u = e^x, du = e^x dx$.

   $$\int \frac{1}{1+u^2}\,du = \arctan(u) + C = \arctan(e^x) + C$$

   Check: $\frac{d}{dx}(\arctan(e^x)) = \frac{1}{1+(e^x)^2} \cdot e^x = \frac{e^x}{1+e^{2x}}$.

6. $\displaystyle\int_1^e \frac{(\ln x)^3}{x}\,dx$

u-sub.
Let $u = \ln x, du = \frac{1}{x}dx$. Limits: $x = 1 \to u = 0, x = e \to u = 1$.

$$\int_0^1 u^3\,du = \frac{1}{4}u^4\Big|_0^1 = \frac{1}{4}$$

7. $\displaystyle\int \frac{x+2}{x^2+1}\,dx$

Split the integral.
$\displaystyle\int \frac{x}{x^2+1}\,dx + \int \frac{2}{x^2+1}\,dx$. First part uses $u = x^2+1, du = 2xdx$. Second part is arctan.

$$\frac{1}{2}\ln(x^2+1) + 2\arctan x + C$$

Check: $\frac{d}{dx}(\frac{1}{2}\ln(x^2+1) + 2\arctan x) = \frac{x}{x^2+1} + \frac{2}{x^2+1} = \frac{x+2}{x^2+1}$.

8. $\displaystyle\int t^2 \ln t\,dt$

Integration by Parts.
$u = \ln t \qquad dv = t^2 dt$
$du = \frac{1}{t}dt \qquad v = \frac{1}{3}t^3$

$$\frac{1}{3}t^3\ln t - \int \frac{1}{3}t^2\,dt = \frac{1}{3}t^3\ln t - \frac{1}{9}t^3 + C$$

Check: $\frac{d}{dt}(\frac{1}{3}t^3\ln t - \frac{1}{9}t^3) = t^2\ln t + \frac{1}{3}t^2 - \frac{1}{3}t^2 = t^2\ln t$.

9. $\displaystyle\int \frac{1}{x\ln x}\,dx$

u-sub.
Let $u = \ln x, du = \frac{1}{x}dx$.

$$\int \frac{1}{u}\,du = \ln|u| + C = \ln|\ln x| + C$$

Check: $\frac{d}{dx}(\ln(\ln x)) = \frac{1}{\ln x}\cdot\frac{1}{x} = \frac{1}{x\ln x}$.

10. $\displaystyle\int_0^{\pi/4} \tan\theta\sec^2\theta\,d\theta$

u-sub.
Let $u = \tan\theta, du = \sec^2\theta d\theta$. Limits: $\theta = 0 \to u = 0, \theta = \pi/4 \to u = 1$.

$$\int_0^1 u\,du = \frac{1}{2}u^2\Big|_0^1 = \frac{1}{2}$$

11. $\int x^2 e^x \, dx$

Integration by Parts (Two Steps).

Step 1: $u = x^2$ $dv = e^x dx$; $du = 2x dx$ $v = e^x$

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx$$

Step 2: $u = 2x$ $dv = e^x dx$; $du = 2dx$ $v = e^x$

$$\int x^2 e^x \, dx = x^2 e^x - (2x e^x - \int 2e^x \, dx) = x^2 e^x - 2x e^x + 2e^x + C$$

Check: $\frac{d}{dx}(x^2 e^x - 2x e^x + 2e^x) = (2x e^x + x^2 e^x) - (2e^x + 2x e^x) + 2e^x = x^2 e^x$.

12. $\int \frac{1}{x^2 + 4} \, dx$

We want this to be something like $\frac{1}{1+x^2}$ (note the 1!)

So factor out a 4 from the denominator to get $\frac{1}{4(\frac{x^2}{4}+1)}$.

Inverse Trig (u-sub with $u = \frac{x}{2}$) (since $\left(\frac{x}{2}\right)^2 = x^2/4$).

$\frac{1}{4}\int \frac{1}{(x/2)^2+1} \, dx = \frac{1}{2}\arctan(x/2) + C$. Check: $\frac{d}{dx}(\frac{1}{2}\arctan(x/2)) = \frac{1}{2} \cdot \frac{1}{1+(x/2)^2} \cdot \frac{1}{2} = \frac{1}{4+x^2}$.

13. $\int \frac{\ln x}{x^2} \, dx$

Integration by Parts.

$u = \ln x$ $dv = x^{-2} dx$; $du = \frac{1}{x} dx$ $v = -x^{-1}$

$$-\frac{\ln x}{x} + \int x^{-2} \, dx = -\frac{\ln x}{x} - \frac{1}{x} + C$$

Check: $\frac{d}{dx}(-\frac{\ln x}{x} - \frac{1}{x}) = -(\frac{1-\ln x}{x^2}) + \frac{1}{x^2} = \frac{\ln x}{x^2}$.

14. $\int x^3 \sqrt{x^4 + 1} \, dx$

u-sub.

Let $u = x^4 + 1, du = 4x^3 dx$.

$$\frac{1}{4}\int \sqrt{u} \, du = \frac{1}{4} \cdot \frac{2}{3} u^{3/2} + C = \frac{1}{6}(x^4+1)^{3/2} + C$$

Check: $\frac{d}{dx}(\frac{1}{6}(x^4+1)^{3/2}) = \frac{1}{6} \cdot \frac{3}{2}(x^4+1)^{1/2} \cdot 4x^3 = x^3\sqrt{x^4+1}$.

15. $\int \sin(2x)\cos(2x)\,dx$

u-sub.
Let $u = \sin(2x), du = 2\cos(2x)dx$.

$$\frac{1}{2}\int u\,du = \frac{1}{4}u^2 + C = \frac{1}{4}\sin^2(2x) + C$$

Check: $\frac{d}{dx}(\frac{1}{4}\sin^2(2x)) = \frac{1}{4}\cdot 2\sin(2x)\cdot 2\cos(2x) = \sin(2x)\cos(2x)$.

16. $\int x\ln(x+1)\,dx$

Integration by Parts followed by u-substitution.

$u = \ln(x+1)$ $\quad dv = xdx$
$du = \frac{1}{x+1}dx$ $\quad v = \frac{1}{2}x^2$

Using the integration by parts formula:

$$\int x\ln(x+1)\,dx = \frac{1}{2}x^2\ln(x+1) - \int \frac{x^2}{2(x+1)}\,dx$$

To evaluate the remaining integral, we use the substitution $w = x+1$, which implies $x = w-1$ and $dx = dw$:

$$\frac{1}{2}\int \frac{(w-1)^2}{w}\,dw = \frac{1}{2}\int \frac{w^2-2w+1}{w}\,dw$$

$$= \frac{1}{2}\int \left(w - 2 + \frac{1}{w}\right)\,dw = \frac{1}{2}\left(\frac{1}{2}w^2 - 2w + \ln|w|\right) + C$$

Substituting $w = x+1$ back in:

$$= \frac{1}{4}(x+1)^2 - (x+1) + \frac{1}{2}\ln|x+1| + C$$

Combining this with the first part of the integration by parts (and absorbing constants):

$$\int x\ln(x+1)\,dx = \frac{1}{2}x^2\ln(x+1) - \left[\frac{1}{4}(x+1)^2 - (x+1) + \frac{1}{2}\ln|x+1|\right] + C$$

$$= \frac{1}{2}x^2\ln(x+1) - \frac{1}{4}x^2 + \frac{1}{2}x + \frac{3}{4} - \frac{1}{2}\ln|x+1| + C$$

Check: $\frac{d}{dx}\left(\frac{1}{2}x^2\ln(x+1) - \frac{1}{4}x^2 + \frac{1}{2}x - \frac{1}{2}\ln(x+1)\right)$

$$= \left(x\ln(x+1) + \frac{x^2}{2(x+1)}\right) - \frac{1}{2}x + \frac{1}{2} - \frac{1}{2(x+1)}$$

$$= x\ln(x+1) + \frac{x^2-1}{2(x+1)} - \frac{1}{2}x + \frac{1}{2} = x\ln(x+1) + \frac{x-1}{2} - \frac{x-1}{2} = x\ln(x+1).$$

17. A student attempted to evaluate $\int xe^x\,dx$ as follows:

$u = e^x$ $\quad dv = xdx$
$du = e^x dx$ $\quad v = \frac{1}{2}x^2$

$$\int xe^x\,dx = \frac{1}{2}x^2e^x - \int \frac{1}{2}x^2e^x\,dx$$

Identify their error, then provide the correct solution.

Error: The student's choice of u and dv resulted in an integral where the power of x increased ($\int x^2 e^x \, dx$), making the new integral more complicated than the original.

Correct Parts: $u = x$ $\quad dv = e^x dx$
$du = dx$ $\quad v = e^x$

$$\int xe^x \, dx = xe^x - \int e^x \, dx = xe^x - e^x + C$$

Check: $\frac{d}{dx}(xe^x - e^x + C) = (e^x + xe^x) - e^x = xe^x$.

18. A student evaluated $\int (3x+1)^4 \, dx$ as follows:

$$\int (3x+1)^4 \, dx = \frac{1}{5}(3x+1)^5 + C$$

Check: $\frac{d}{dx}(\frac{1}{5}(3x+1)^5 + C) = 3(3x+1)^4$

Identify their error, then provide the correct solution.

Error: The student treated the integral as a simple power rule but forgot to account for the constant coefficient (3) inside the parentheses. Their own check shows the result is off by a factor of 3.

Correct Solution: Let $u = 3x+1$, so $du = 3dx \implies \frac{1}{3}du = dx$.

$$\int (3x+1)^4 \, dx = \frac{1}{3}\int u^4 \, du = \frac{1}{3} \cdot \frac{1}{5}u^5 + C = \frac{1}{15}(3x+1)^5 + C$$

Check: $\frac{d}{dx}(\frac{1}{15}(3x+1)^5 + C) = \frac{1}{15} \cdot 5(3x+1)^4 \cdot 3 = (3x+1)^4$.

19. A student evaluated $\int \cos^2(x) \, dx$ as follows:

$\cos^2(x) = 1 - \sin^2(x)$

$$\text{So } \int \cos^2(x) \, dx = \int 1 - \sin^2(x) \, dx = x - \frac{1}{3}\sin^3(x) + C$$

Identify their error, then provide the correct solution.

Error: The student attempted to use the power rule on $\sin^2(x)$ without the necessary $\cos(x)dx$ term required to satisfy a u-substitution.

Correct Solution: Use the trig identity $\cos^2 x = \frac{1}{2}(\cos(2x) + 1)$.

$$\int \cos^2 x \, dx = \frac{1}{2}\int (\cos(2x) + 1) \, dx = \frac{1}{4}\sin(2x) + \frac{1}{2}x + C$$

Check: $\frac{d}{dx}(\frac{1}{4}\sin(2x) + \frac{1}{2}x) = \frac{1}{2}\cos(2x) + \frac{1}{2} = \cos^2 x$.

20. A student evaluated $\int x\cos(x)\,dx$ as follows:

$$\begin{array}{ll} u = x & dv = \cos(x)dx \\ du = dx & v = \sin(x) \end{array}$$

$$\int x\cos(x)\,dx = x\sin(x) + \int \sin(x)\,dx = x\sin(x) - \cos(x) + C$$

Identify their error, then provide the correct result.

Error: The student utilized an incorrect formula for integration by parts; the standard algorithm requires the subtraction of the integral of $v\,du$.

Correct Solution:

$$\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x - (-\cos x) + C = x\sin x + \cos x + C$$

Check: $\frac{d}{dx}(x\sin x + \cos x + C) = (\sin x + x\cos x) - \sin x = x\cos x$.